

Yuan Jack Yao

404-528-6519 | yyao411@gatech.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Georgia Institute of Technology

Aug 2025 – Dec 2026

M.Sc. Analytics @ School of Computational Science and Engineering

Atlanta, Georgia

- GPA 4.0/4.0: **Conversational AI, Natural Language**, Machine Learning, ML in Computational Biology

National Taiwan University

Sep 2014 – Jun 2018

Bachelor's Degree in Economics

Taipei, Taiwan

WORK EXPERIENCES

Walmart

Jun 2026 – Aug 2026

Data Scientist III Intern @ Sam's Club AI Acceleration Team

Bentonville, Arkansas

- Deployed an **hourly CronJob** with **Docker** and **Kubernetes**, integrating **asynchronous processing** that reduced **50%** execution time for capturing **1,245 camera snapshots** into **Azure**, using 7-day TTL to save **75%** storage cost
- Extended the **same CronJob pipeline** with a container to **synchronize** Sam's Club zone location data from CSV into **PostgreSQL**, enabling **snapshot content to be mapped onto in-store zone layouts**

SUEZ

May 2022 – Sep 2024

Data Scientist @ Smart and Environmental Solutions (SES) China

Shanghai, China

- Streamlined **NLP-based Keras** classifier for **imbalanced** SCADA alarms, reaching **82% production accuracy**
- Designed a **DBSCAN-based anomaly detector** for air compressors **predictive maintenance**, achieving **70% production F1 score** and forecasting failures **two weeks in advance despite having only a single feature**
- Accelerated clustering on 12 air compressors with Joblib, **reducing runtime by 20% in Linux CronJob**

OPEN SOURCE

LeetCode Wiki (36.3k Stars, 9.4k Forks)

- Authored **8 optimal time and space solutions with tutorials** for **hard** problems
- **No.2454 Next Greater Element IV**: Achieved $O(n)$ time complexity by **double stacks** in 2024, and merged PR in 2026, reducing from original solution's global sorting and ordered set of $O(n \log n)$ time

PROJECTS

JOB: Jack's Online Blackjack

- Developed a full-stack online Blackjack game with casino-style UI via FastAPI backend & vanilla JavaScript
- Implemented Macau style complex rules: up to 3 splits, double after split, early pay option, and insurance
- Deployed CI/CD pipeline via GitHub Actions and Railway, enabling automatic deployment on every git push

LinesShines

- Developed an NFL metrics platform with multi-team filtering and exportable high resolution visualizations
- Designed draggable player insight panels showing context-aware rank and percentile that update based on filters

SkyHorse

- Authored in-depth tutorials for 30+ LeetCode hard problems, delivering optimized solutions & complexity analysis
- Designed tutorials to cover algorithmic intuition, step-by-step explanations, and Python & C++ implementations
- Implemented entire website supports English and Chinese, allowing users to switch seamlessly

EcoTeller

- Engineered Qwen2.5-3B-Instruct fine-tuning to extract and quantify environmental issues from news titles
- Designed sparse encoding pipeline to convert text vectors into structured prompts for Qwen's chat template

TECHNICAL SKILLS

Gen AI & ML: LLM Fine-tuning, Distillation, Prompt Engineering, PyTorch, Hugging Face, TensorFlow, Keras

CI/CD & Cloud: Linux, GitHub, Bitbucket, Docker, Kubernetes, Azure, AWS, GCP

Programming & Databases: Python, C++, MySQL, PostgreSQL, ETL Pipelines

Web & Visualization: FastAPI, Flask, JavaScript, HTML, CSS, Web Scraping